Robust Waste Material Classification Using Transfer Learning

Do CNNs Learn Materials or Visual Shortcuts?

Course: MSML640 Computer Vision | Team: Haider Khan and Batamgalan Enkhtaivan

Abstract

This project studies how well a pretrained convolutional neural network can classify recyclable waste materials under realistic image conditions. We built a four-class image classifier for cardboard, glass, metal, and plastic using a 4-class subset of the RealWaste dataset. The model uses an ImageNet-pretrained EfficientNet-B0 backbone with a new classifier head for the four waste categories.

The goal was not only to achieve high accuracy, but to understand whether the model learns meaningful material features or relies on visual shortcuts such as color, lighting, shine, transparency, object shape, and background. To test this, we compared four training configurations: baseline, augmentation, synthetic data, and synthetic data plus augmentation. We also performed robustness testing under blur, noise, dark lighting, bright lighting, and occlusion.

The best model was the synthetic-data configuration, which achieved 73.33% test accuracy compared with 70.00% for the baseline. This suggests that controlled synthetic perturbations helped the model generalize slightly better. However, the model still struggled with glass, plastic, and metal because these materials often share visual properties such as reflection, transparency, and shine.

1. Problem Statement

Waste sorting is a practical computer vision problem. In recycling systems, incorrect sorting of plastic, glass, cardboard, and metal can reduce recycling efficiency and increase contamination. A simple image classifier could help users, recycling centers, or smart bins identify the likely material category from a photo.

However, material classification is not always straightforward. Glass and plastic can both be transparent. Metal and plastic can both appear shiny. Cardboard can sometimes be confused with paper-like or colorful packaging. Because of this, our project asks whether a pretrained CNN learns meaningful material features or relies on visual shortcuts.

2. Dataset

We used a 4-class subset of the RealWaste image classification dataset: cardboard, glass, metal, and plastic. The original selected class counts were Cardboard: 461, Glass: 420, Metal: 790, and Plastic: 921 images.

Split	Images per Class	Total Images
Training	70	280
Validation	15	60
Test	15	60

Synthetic images were created only from the training set. Validation and test data remained real images only, which avoids data leakage.

3. Model Selection

We used EfficientNet-B0 pretrained on ImageNet as the base model. EfficientNet-B0 was chosen because it is lightweight, efficient, and suitable for small-to-medium image classification tasks. Since our dataset is

relatively small, transfer learning helps the model start with useful visual features such as edges, shapes, textures, and object patterns.

The feature extractor was frozen, and the final classifier layer was replaced with a new four-class classification head.

4. Experimental Design

Config	Name	Description
1	Baseline	Real training data only
2	Augmentation	Real training data with crop, flip, rotation, and color jitter
3	Synthetic	Real training data plus synthetic training images
4	Synthetic + Augmented	Real + synthetic data with augmentation

Synthetic images were generated using controlled perturbations such as blur, brightness changes, contrast changes, rotation, and occlusion. These transformations simulate real-world conditions such as motion blur, poor lighting, partial object blockage, and imperfect framing.

5. Evaluation Methods

We evaluated the models using training and validation curves, final test accuracy, confusion matrices, baseline-vs-best comparison, robustness testing, and qualitative error examples. The test set contained 60 images total, with 15 images per class. Because each test image changes accuracy by about 1.67%, we interpret small accuracy differences carefully.

6. Main Results

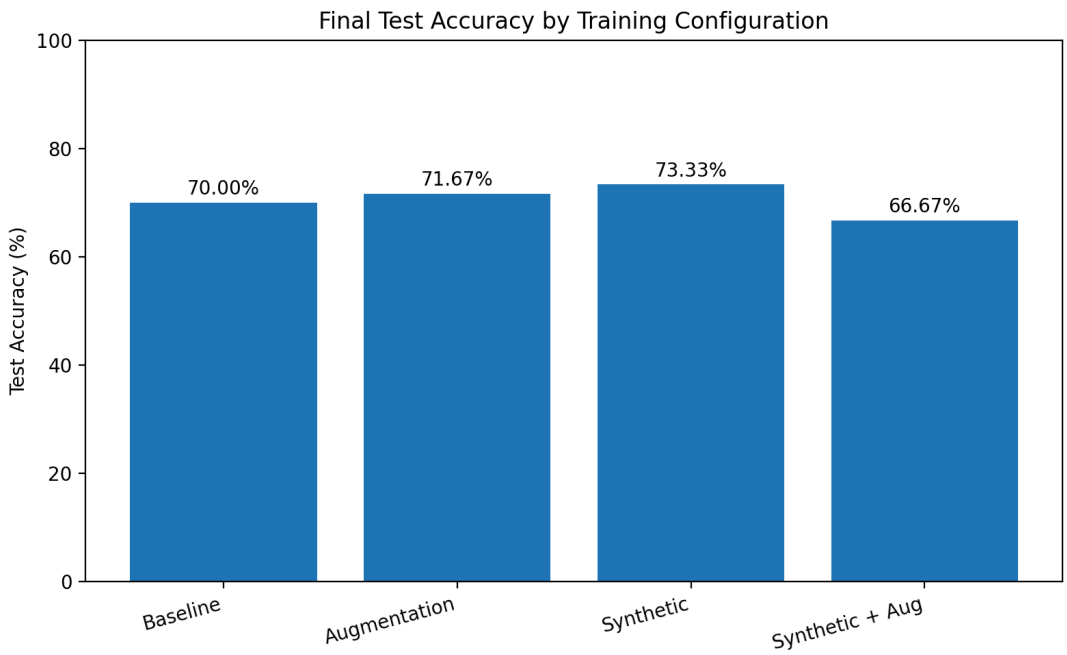

Figure 1. Final test accuracy by training configuration.

Configuration	Test Accuracy

Baseline	70.00%
Augmentation	71.67%
Synthetic	73.33%
Synthetic + Augmented	66.67%

The synthetic-data model achieved the best test accuracy at 73.33%, improving over the baseline by 3.33 percentage points. The improvement is modest, but it supports the idea that controlled synthetic perturbations can help the model generalize slightly better than clean real images alone. The synthetic + augmented configuration performed worse, suggesting that too many transformations may distort useful material cues in a small dataset.

7. Training Curve Analysis

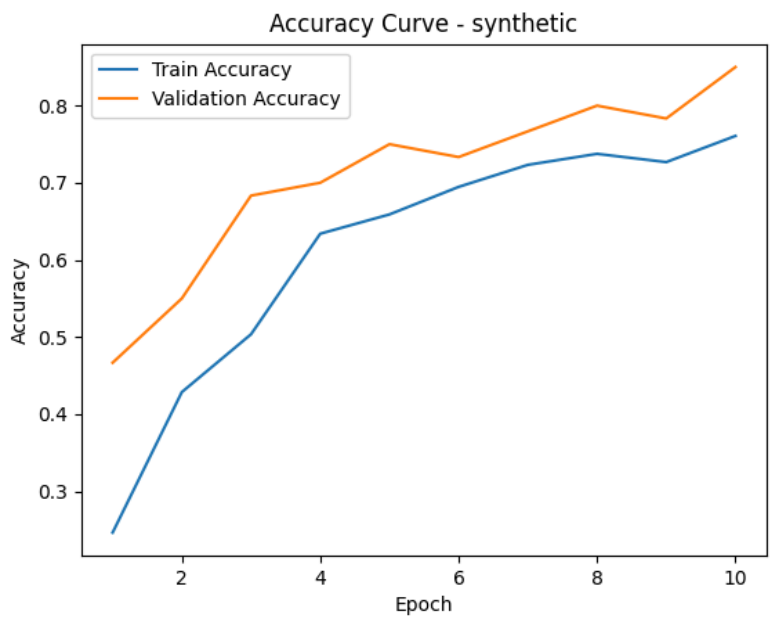

Figure 2. Synthetic model training and validation accuracy curve.

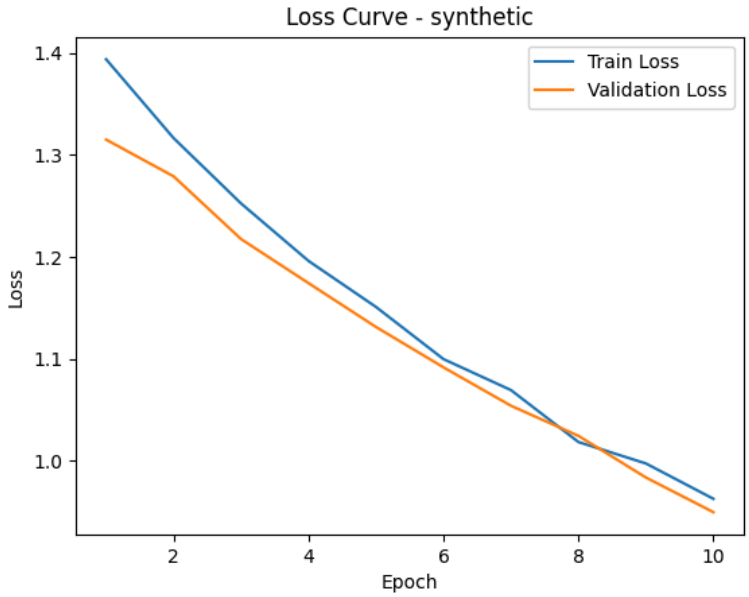

Figure 3. Synthetic model training and validation loss curve.

Across all configurations, training and validation loss generally decreased over 10 epochs. The synthetic model showed the strongest validation accuracy trend, reaching 81.67% validation accuracy by the end of training. This was higher than the baseline validation accuracy of 68.33%.

8. Confusion Matrix Analysis

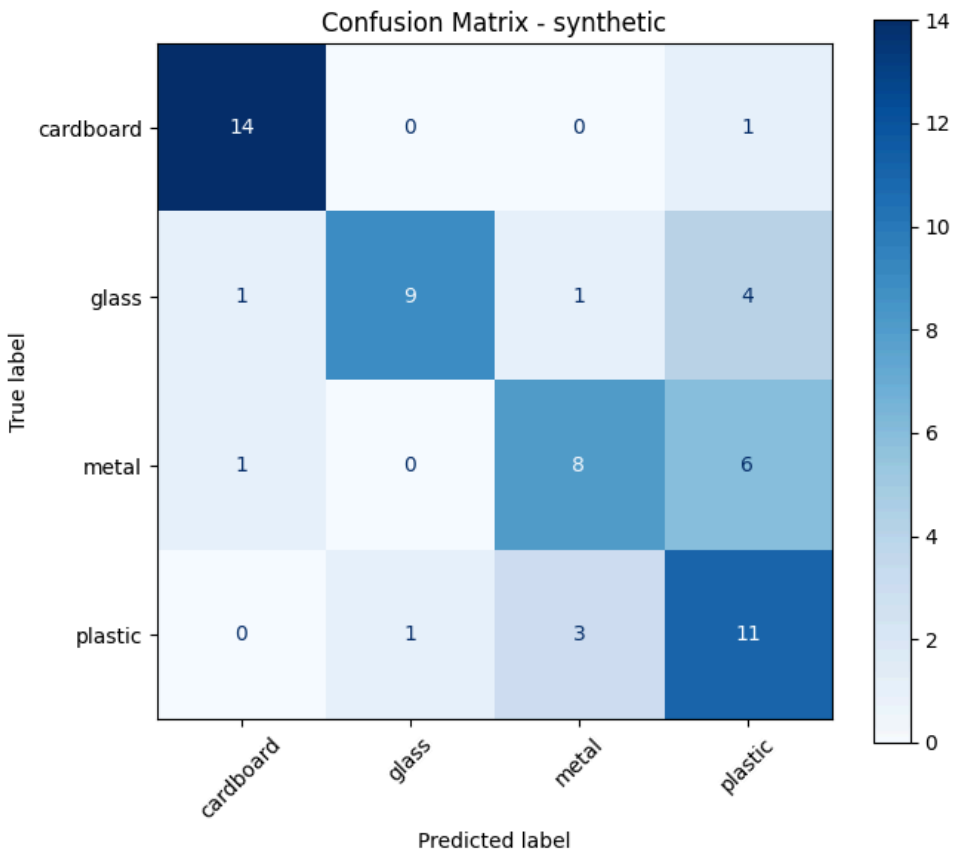

Figure 4. Confusion matrix for the synthetic-data model.

The confusion matrices show that cardboard was the easiest class across most configurations. This is likely because cardboard has a more consistent texture, color, and box-like appearance. The difficult classes were glass, metal, and plastic because they share visual properties such as transparency, shine, reflection, and irregular packaging shape.

For the best model, the synthetic-data model, the confusion matrix showed Cardboard: 14/15 correct, Glass: 10/15 correct, Metal: 8/15 correct, and Plastic: 12/15 correct. Metal had the most confusion, often being predicted as plastic.

9. Baseline vs Best Model

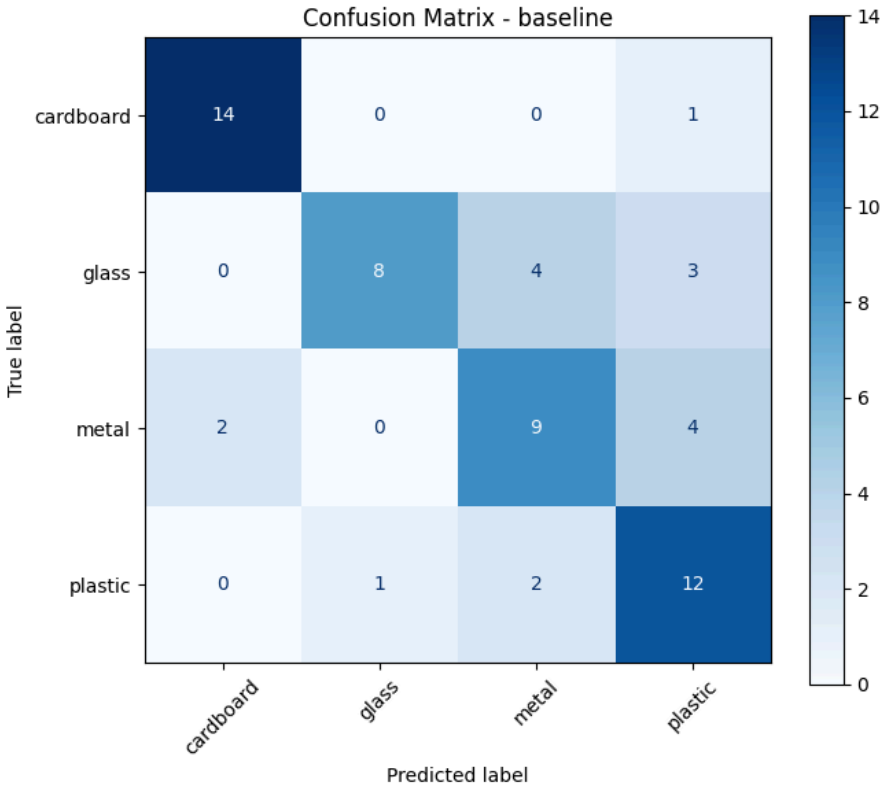

Figure 5. Baseline confusion matrix.

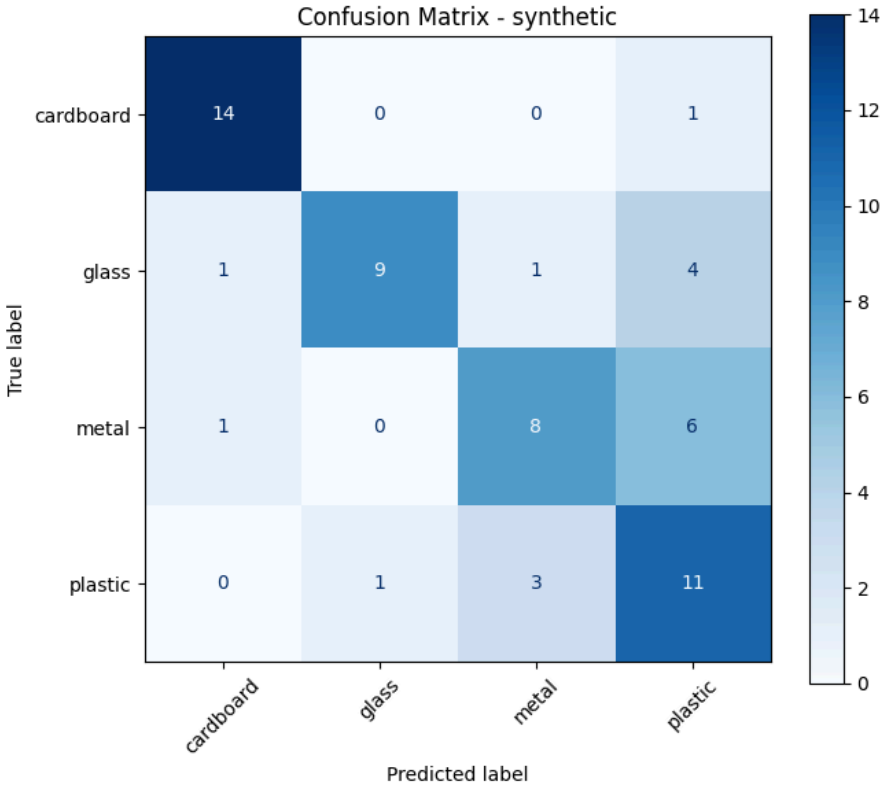

Figure 6. Synthetic-data model confusion matrix.

The baseline model achieved 70.0% test accuracy, while the synthetic-data model achieved 73.33%. The baseline model correctly classified 42 out of 60 test images. The synthetic model correctly classified 44 out of 60 test images. Since the test set is small, this is only two additional correct predictions, so we interpret the improvement carefully.

The synthetic data helped generalization slightly by exposing the model to controlled visual changes during training. However, it did not completely solve confusion between glass, plastic, and metal. This supports our original research question: the model learns useful patterns, but may still rely on visual shortcuts such as shine, transparency, brightness, and object shape.

10. Robustness Testing

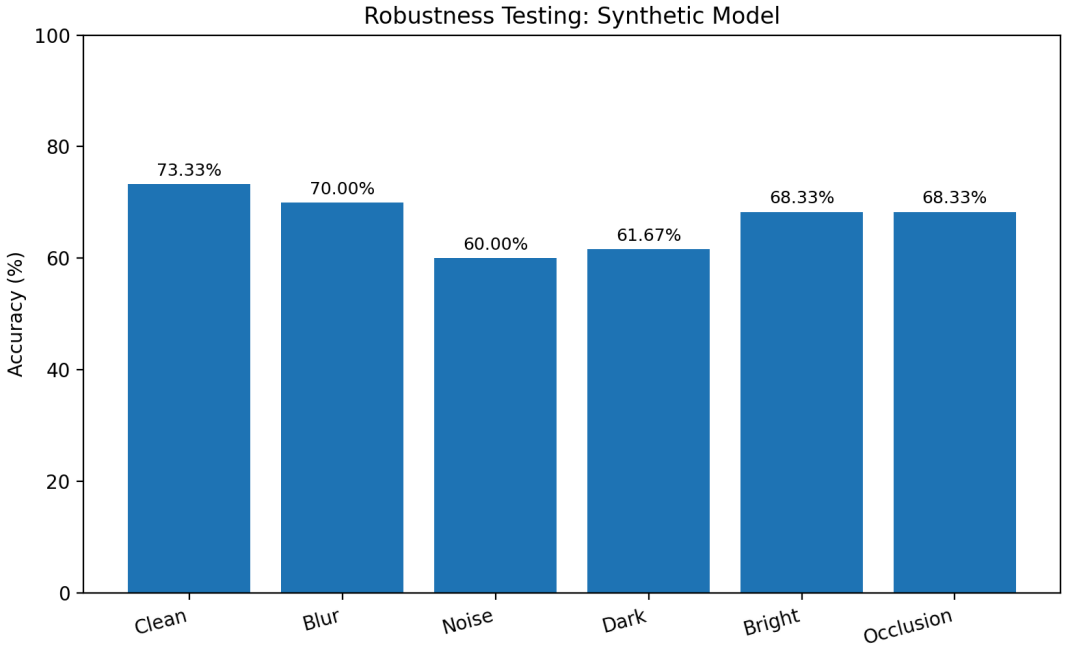

Figure 7. Robustness testing results for the synthetic-data model.

Perturbation	Accuracy
Clean	73.33%
Blur	70.00%
Noise	60.00%
Dark lighting	61.67%
Bright lighting	68.33%
Occlusion	68.33%

The model remained fairly stable under blur, dropping from 73.33% to 70.00%. It also handled bright lighting and occlusion reasonably well. The largest drops occurred under noise and dark lighting, suggesting sensitivity to low-quality camera conditions.

11. Qualitative Error Analysis

Figure 8. Example of cardboard predicted as plastic.

Figure 9. Example of glass predicted as another class.

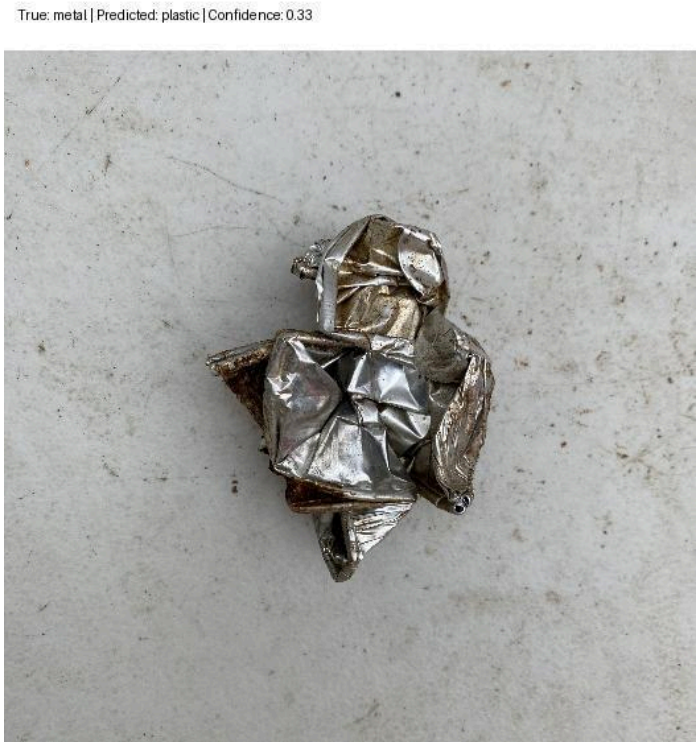

Figure 10. Example of metal predicted as plastic.

Figure 11. Example of plastic predicted as metal.

The saved misclassified examples show several failure patterns. Some plastic containers were predicted as metal when they had dark, reflective, or shiny surfaces. Some crumpled metal objects were predicted as plastic because the shape looked like packaging. Some glass bottles were misclassified when labels or background dominated the image. These errors suggest that the model partly relies on visual shortcuts rather than true material identity.

12. Preprocessing Ablation

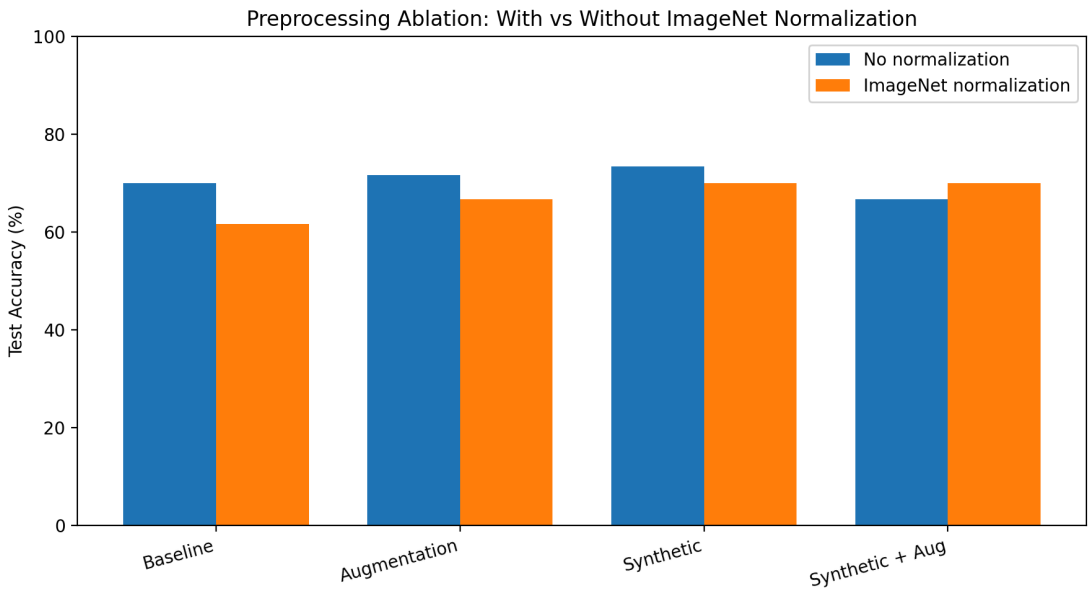

Figure 12. Preprocessing ablation: no normalization vs ImageNet normalization.

Preprocessing	Baseline	Augmentation	Synthetic	Synthetic + Augmented
No normalization	70.00%	71.67%	73.33%	66.67%
ImageNet normalization	61.67%	66.67%	70.00%	70.00%

Although ImageNet normalization is commonly used with pretrained models, the no-normalization setting performed better in this frozen-backbone experiment. We therefore used the no-normalization setting for the final reported results and treated the comparison as a preprocessing ablation.

13. Limitations

The dataset subset is small, and the test set contains only 60 images. The frozen EfficientNet-B0 backbone may limit the model's ability to learn waste-specific material features. Synthetic data was created using simple perturbations, which do not fully represent the range of real-world waste images. Also, a single RGB image may not always contain enough information to perfectly determine material identity.

14. Future Work

Future work could use a larger dataset, fine-tune more layers of EfficientNet-B0, generate more realistic synthetic images, add confidence calibration, and compare additional backbones such as ResNet-50 or MobileNetV2.

15. Tools and Implementation

The project was implemented in Python using PyTorch, torchvision, PIL, NumPy, scikit-learn, and Matplotlib. The repository includes scripts for data preparation, synthetic generation, training, evaluation, robustness testing, and error example collection. All plots and visualizations are supported by source code in the GitHub repository.

16. Conclusion

This project shows that transfer learning can classify waste material images reasonably well even with a small dataset. The best model, trained with synthetic data, achieved 73.33% test accuracy, improving over the 70.00% baseline. This suggests that controlled synthetic perturbations can help generalization.

However, the model still struggles with visually similar materials. Glass, plastic, and metal are often confused because they share features such as shine, transparency, reflectiveness, and packaging-like shape. Robustness testing also showed that noise and dark lighting reduce performance, which is important for real-world use. The main lesson is that the model learns useful visual patterns, but it does not perfectly understand material identity and can depend on visual shortcuts.